Momina & Friends
COS 214
Final Project

WHO ARE WE?

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WHAT ARE WE:

- The project focuses on designing and implementing **a Greenhouse Nursery Simulator**, a virtual representation of a plant nursery.
- It captures the beauty, complexity, and structure of real-world nursery operations.
- It **models the interaction between plants, customers, and staff** within a dynamic and evolving environment.
- The project aims to apply **object-oriented programming** (OOP) and Gof design patterns.
- The goal is to create **a flexible, maintainable, and scalable system.**
- The simulator demonstrates how design patterns simplify complex problems, including:
- Managing different plant types
- Handling unique care requirements
- Coordinating staff duties
- Facilitating customer interactions

Functional requirements :

Greenhouse/Garden Area:

Simulates diverse plant growth and life cycles

Manages watering,sunlight,pruning & fertilising needs

Tracks inventory and supports adding new stock

Staff:

Maintains plant health and monitors growth

Manages inventory and assists customers

Performs care tasks like watering and pruning

Customers and Sales Floor:

Browse and purchase available plants

Customise orders with pots or packaging

Interact with staff for advice and assistance

SIMULATION

HOW IT WORKS:

OUR SELF RUNNING SIMULATION AUTONOMOUSLY MANAGES THE GREENHOUSE NURSERY, MONITORING PLANT HEALTH, STAFF ACTIVITY AND CUSTOMER INTERACTIONS TO ENSURE SMOOTH OPERATION.

THE SIMULATION TRACKS KEY METRICS SUCH AS:

PLANT HEALTH – WATER, SUNLIGHT, FERTILIZER AND PRUNING LEVELS

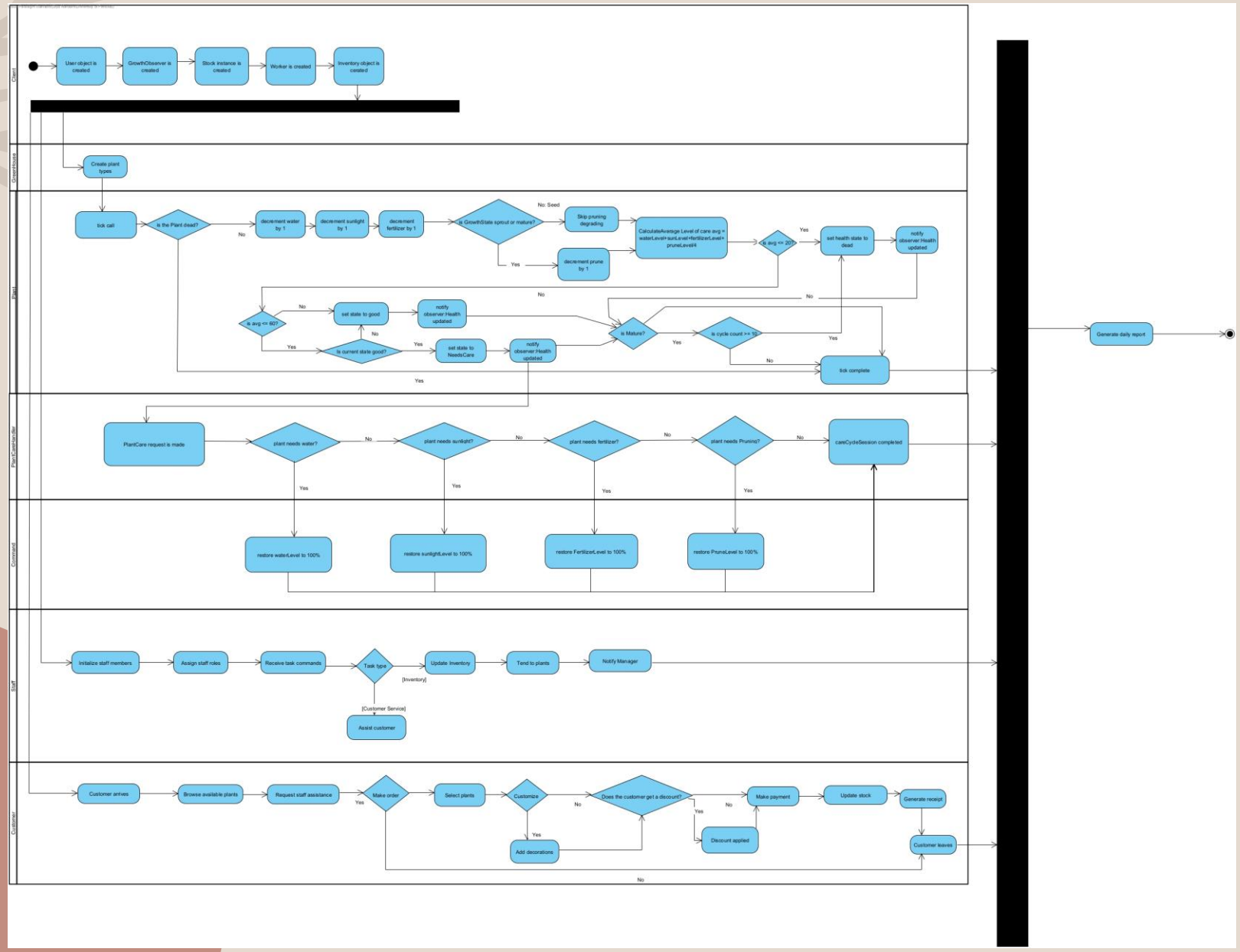
GROWTH PROGRESS – TRACKS EACH PLANTS STAGE FROM SEEDLING TO MATURITY

INVENTORY LEVELS- MONITORS STOCK AVAILABLE FOR SALE OR IN GROWTH

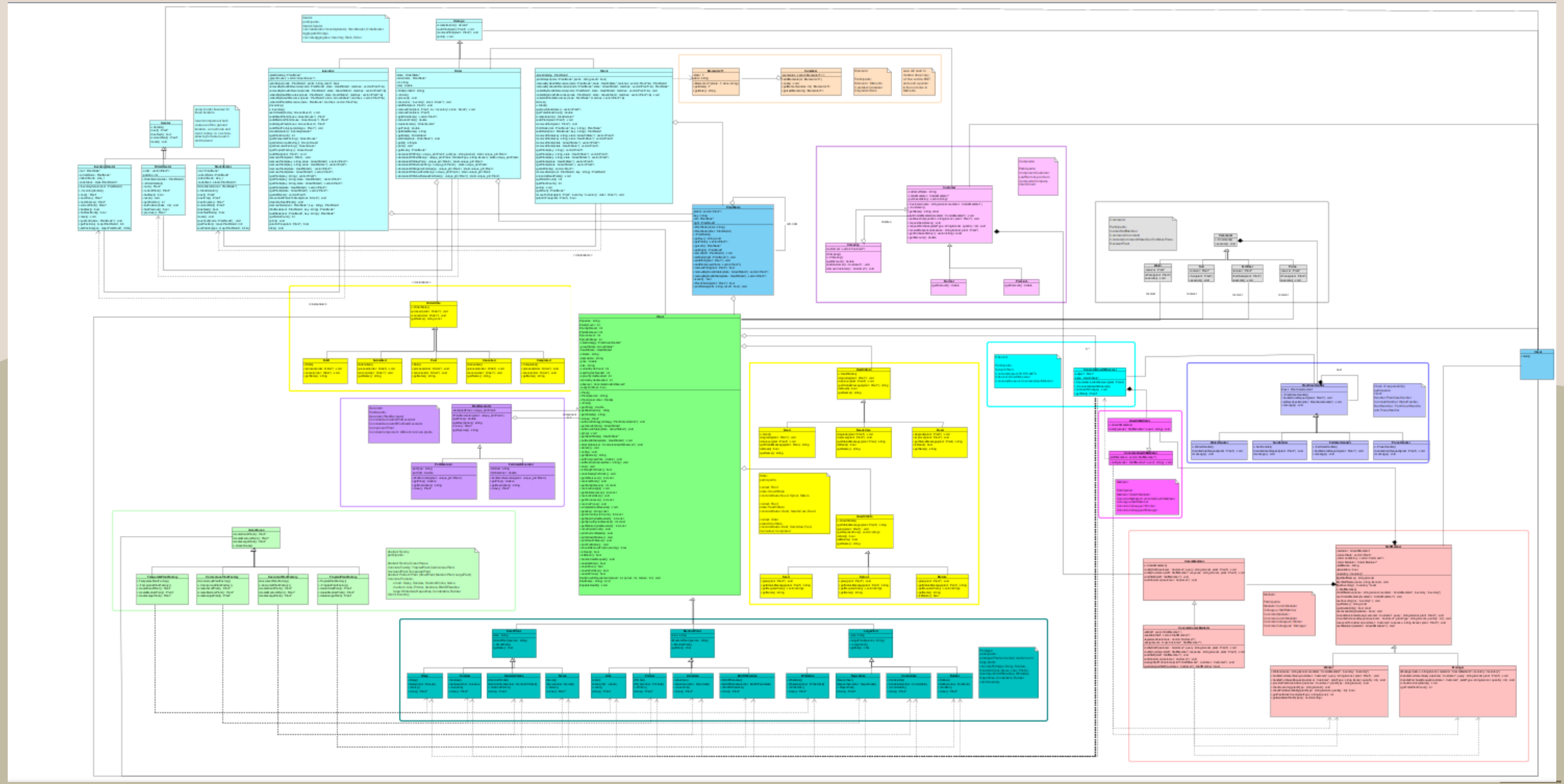
STAFF EFFICIENCY- MEASURES HOW QUICKLY AND EFFECTIVELY STAFF RESPOND TO PLANT NEEDS

CUSTOMER SATISFACTION- REFLECTS HOW WELL CUSTOMER NEEDS ARE MET

Activity Diagram



CLASS DIAGRAM

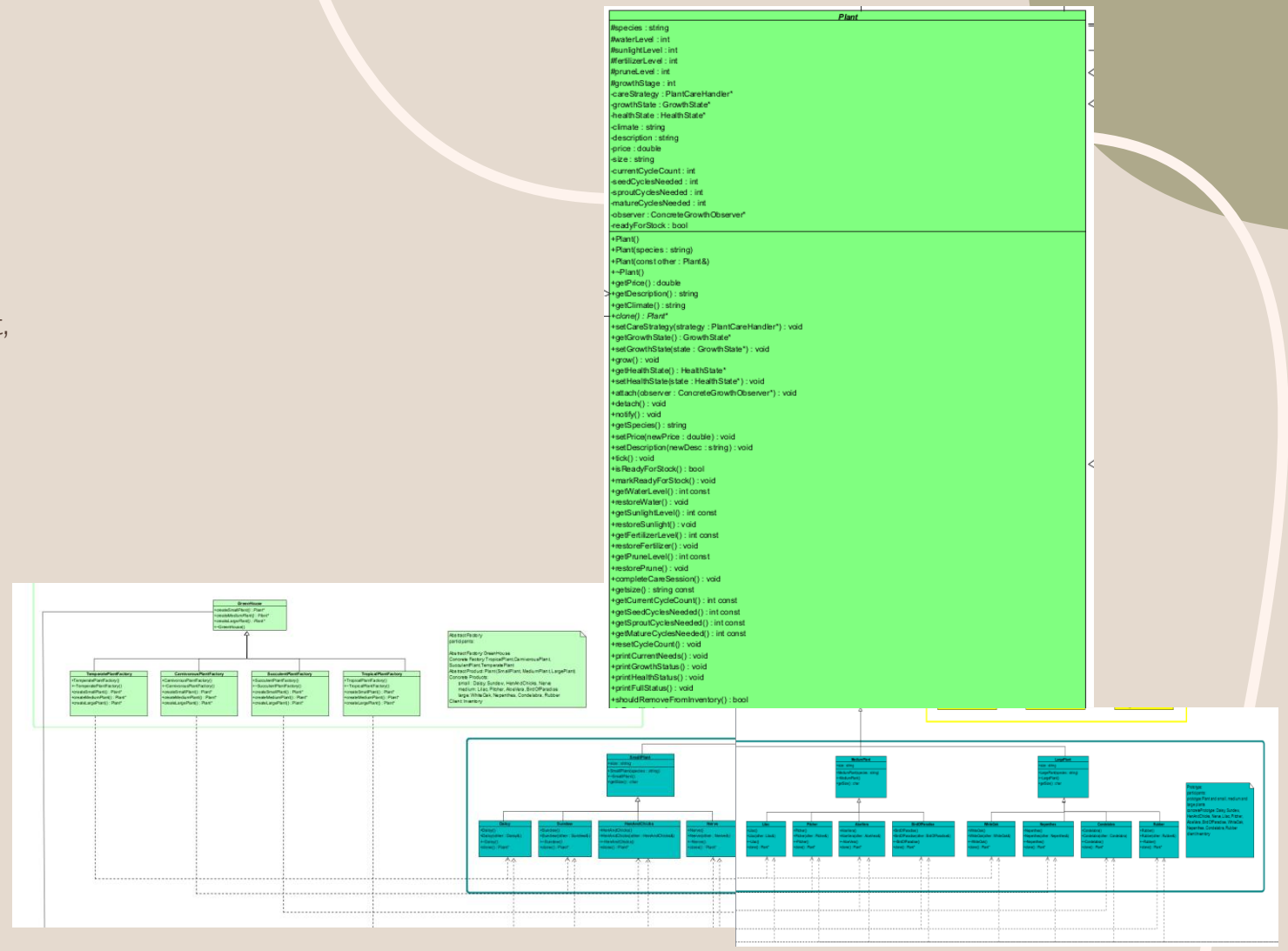


Abstract Factory & Prototype :

Participants:

Abstract Factory: GreenHouse
Concrete Factory: TropicalPlant, CarnivorousPlant, SucculentPlant, TemperatePlant
Abstract Product: Plant(SmallPlant, MediumPlant, LargePlant)
ConcreteProducts:
Small: Daisy,Sundew,HenAndChick,Nerve
Medium:Lilac,Pitcher,AloeVera,BirdOfParadise
Large:WhiteOak,Nepenthes,Condalabra,Rubber
Client:Inventory

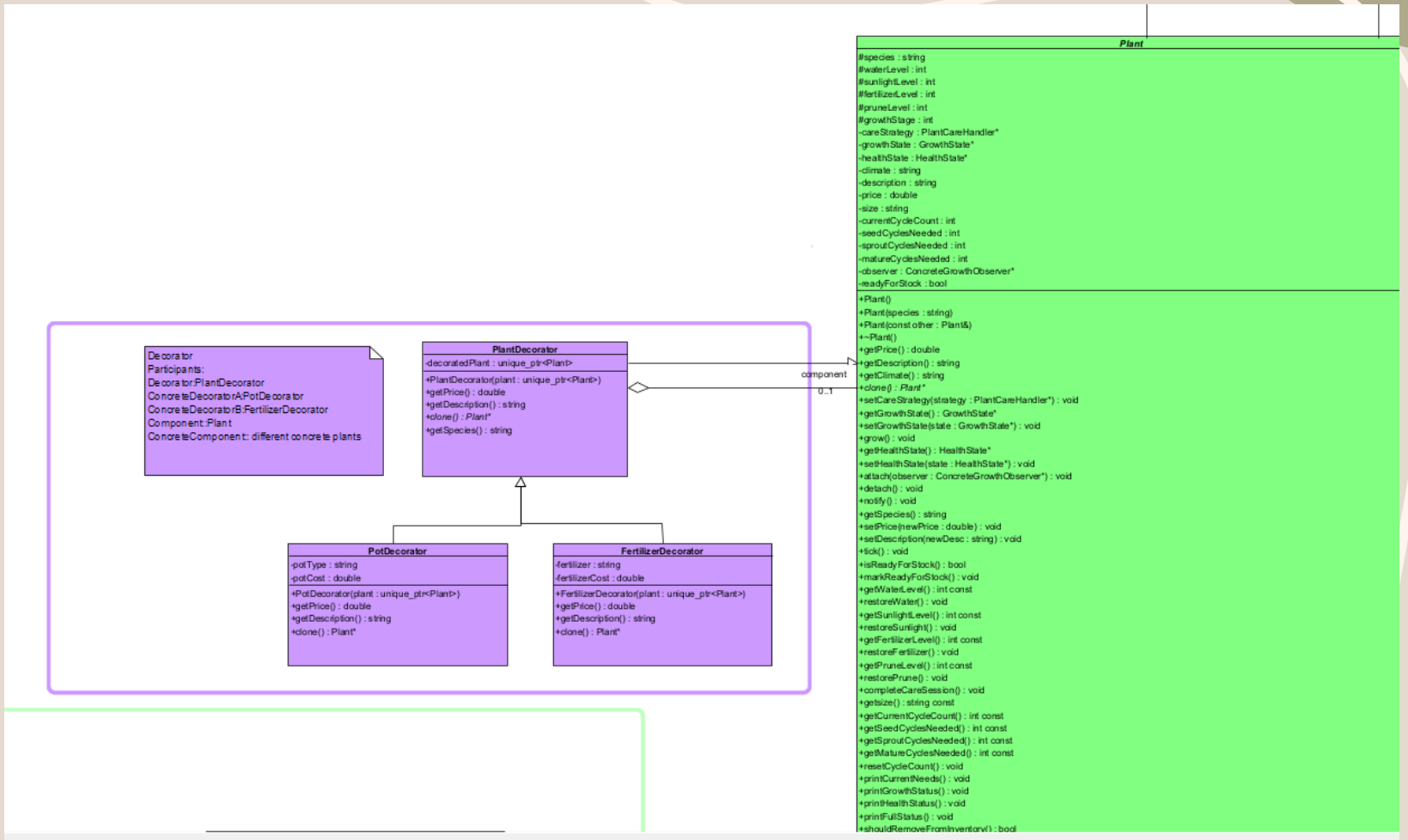
Prototype: Plant and small, medium and large plants
concretePrototype: Daisy, Sundew, HenAndChicks, Nerve, Lilac, Pitcher, AloeVera, BirdOfParadise , WhiteOak, Nepenthes, Condelabra, Rubber
Client: Inventory



DECORATOR :

Participants :

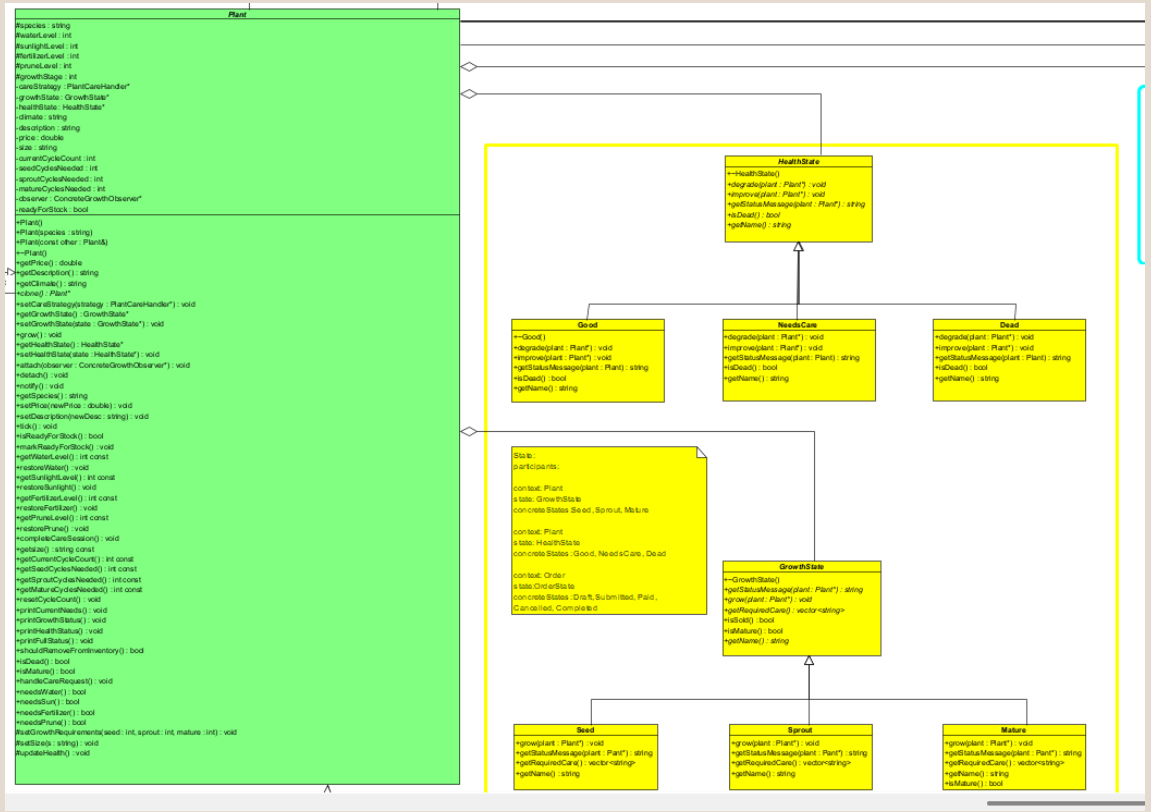
Decorator : PlantDecorator
ConcreteDecorator : PotDecorator & FertilizerDecorator
Component: Plant



State

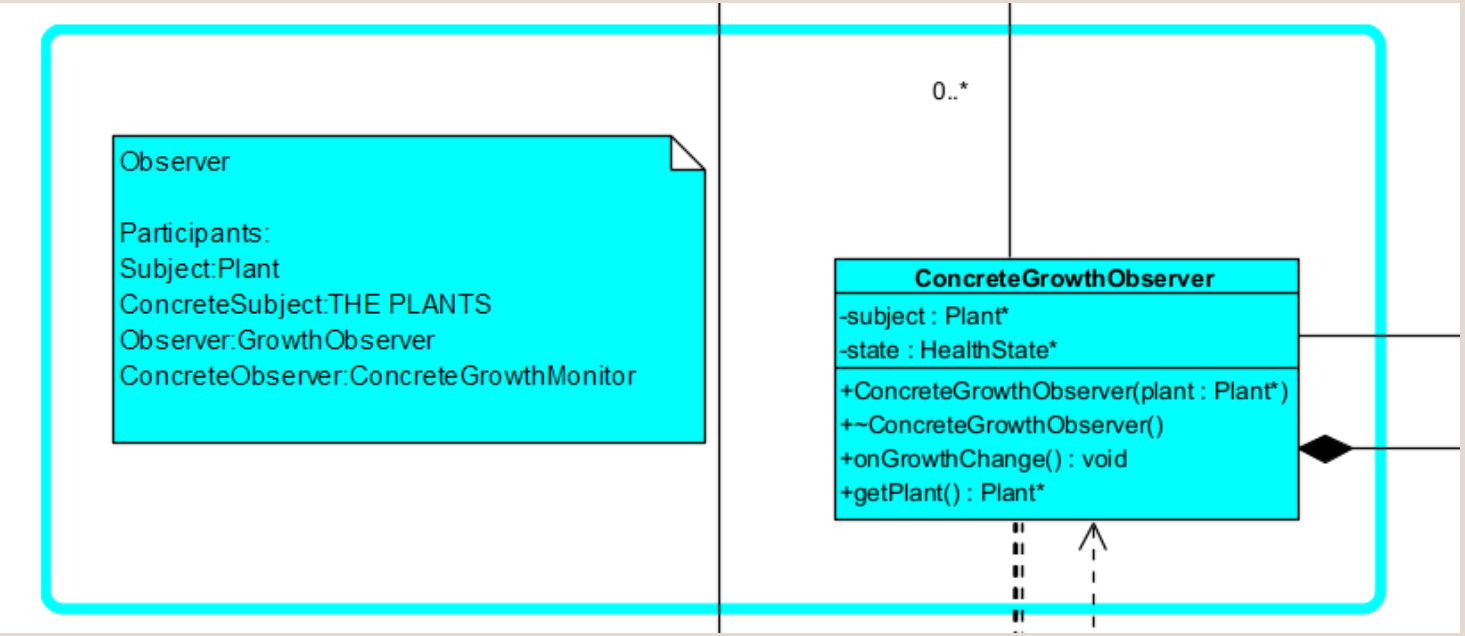
Participants:
Context: Plant
state: GrowthState
concreteStates: Seed, Sprout, Mature

Context:Plant
State: HealthState
concreteStates: Good, NeedsCare, Dead



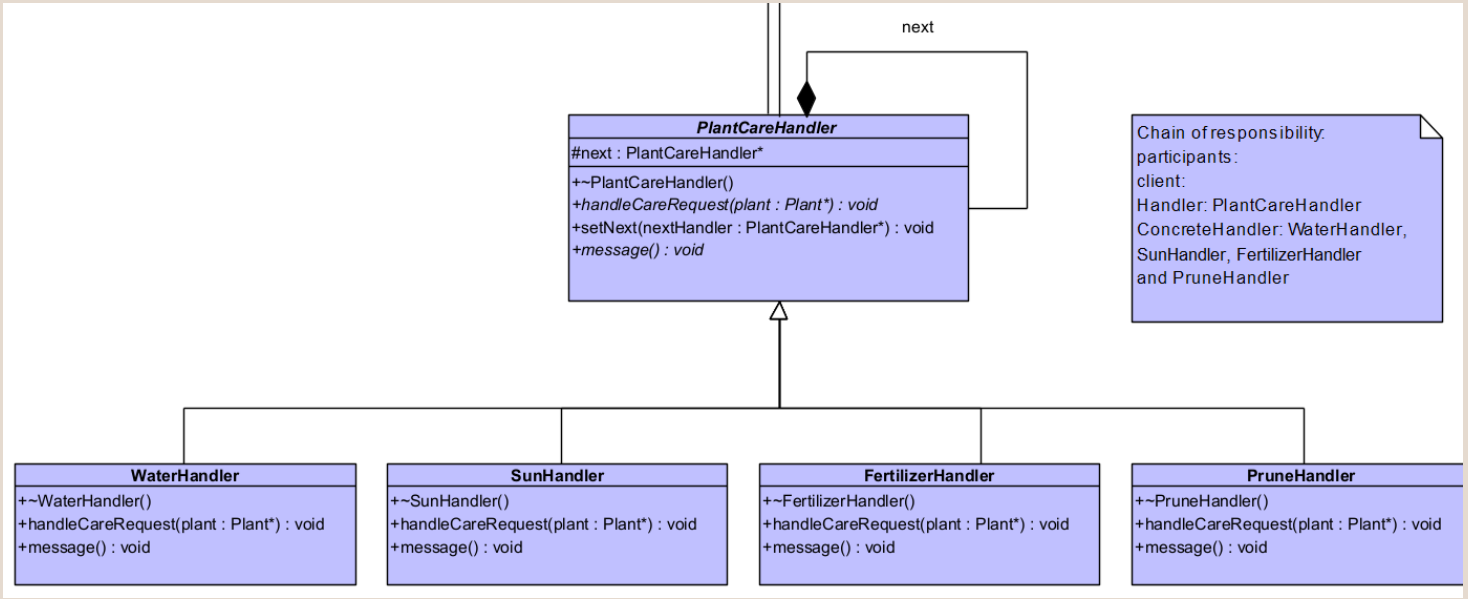
Observer

Participants:
Subject: Plant
ConcreteSubject: The plants
Observer: GrowthObserver
ConcreteObserver:
ConcreteGrowthMonitor



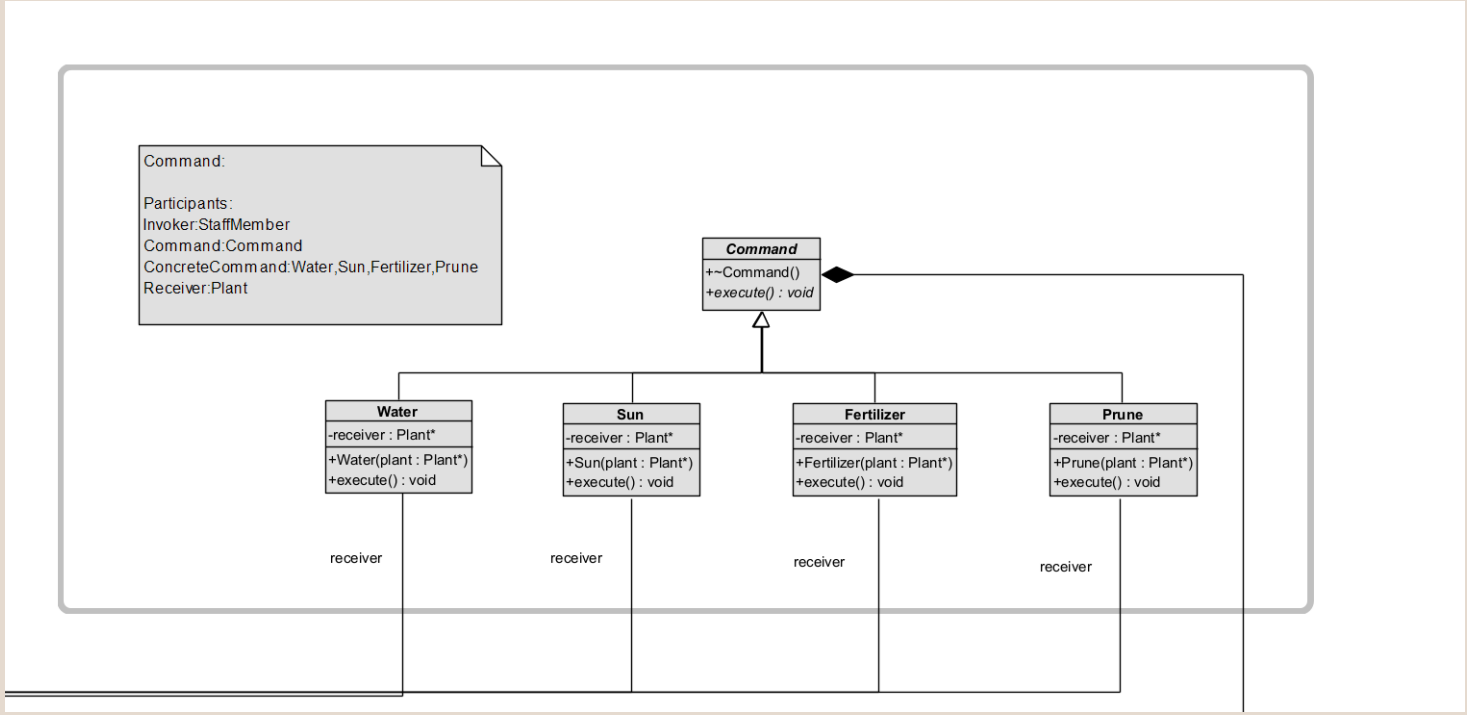
Chain Of Responsibility

Participants:
Handler:
PlantCareHandler
ConcreteHandler:
WaterHandler,
SunHandler,
FertilizerHandler,
PruneHandler



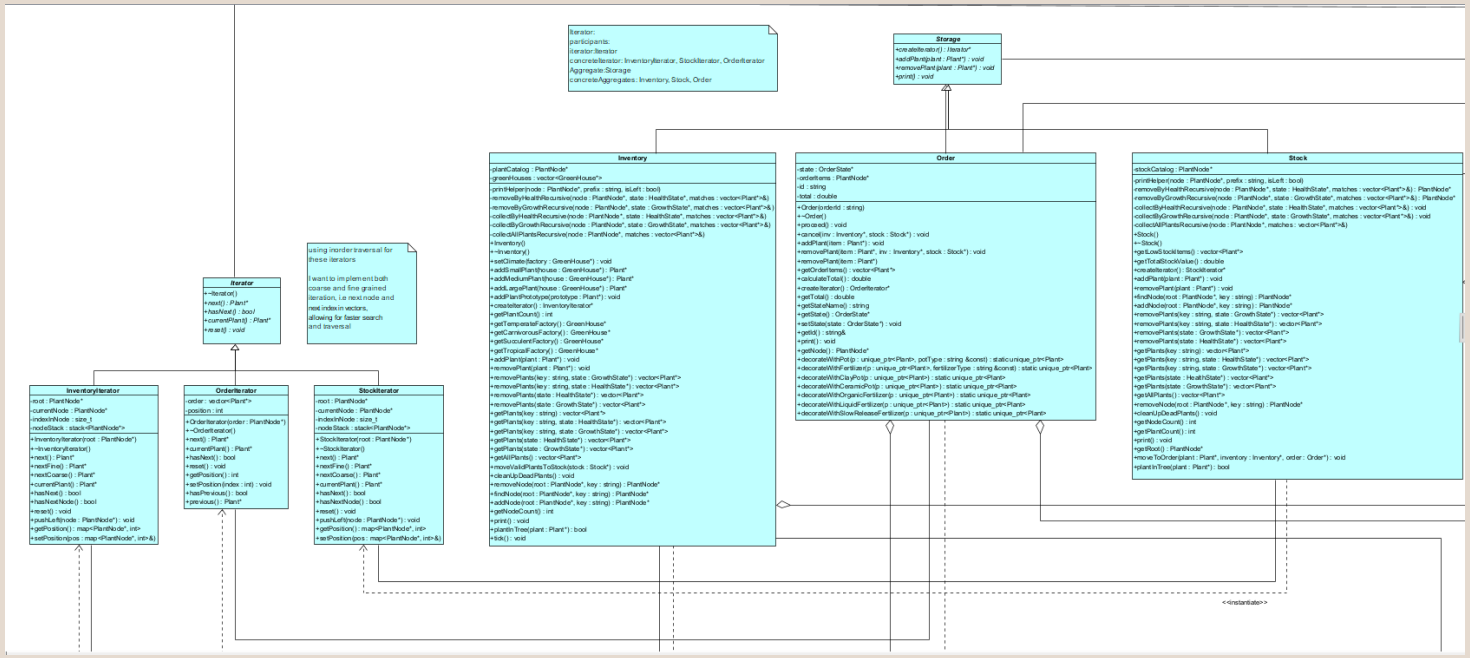
Command

Participants:
Invoker: StaffMember
Command: Command
ConcreteCommand: Water, Sun, Fertilizer, Prune
Receiver: Plant



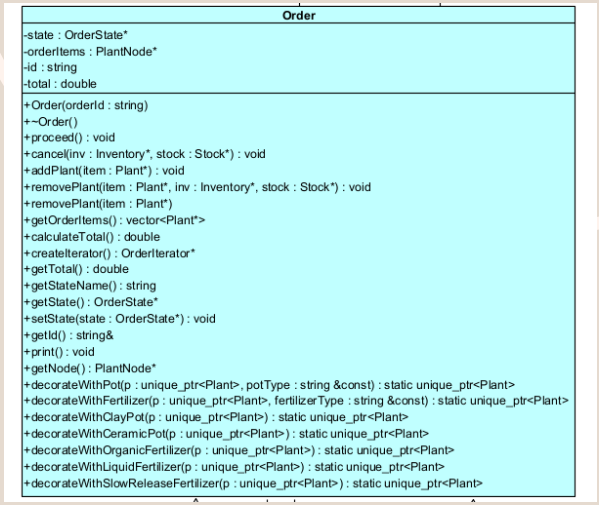
Iterator

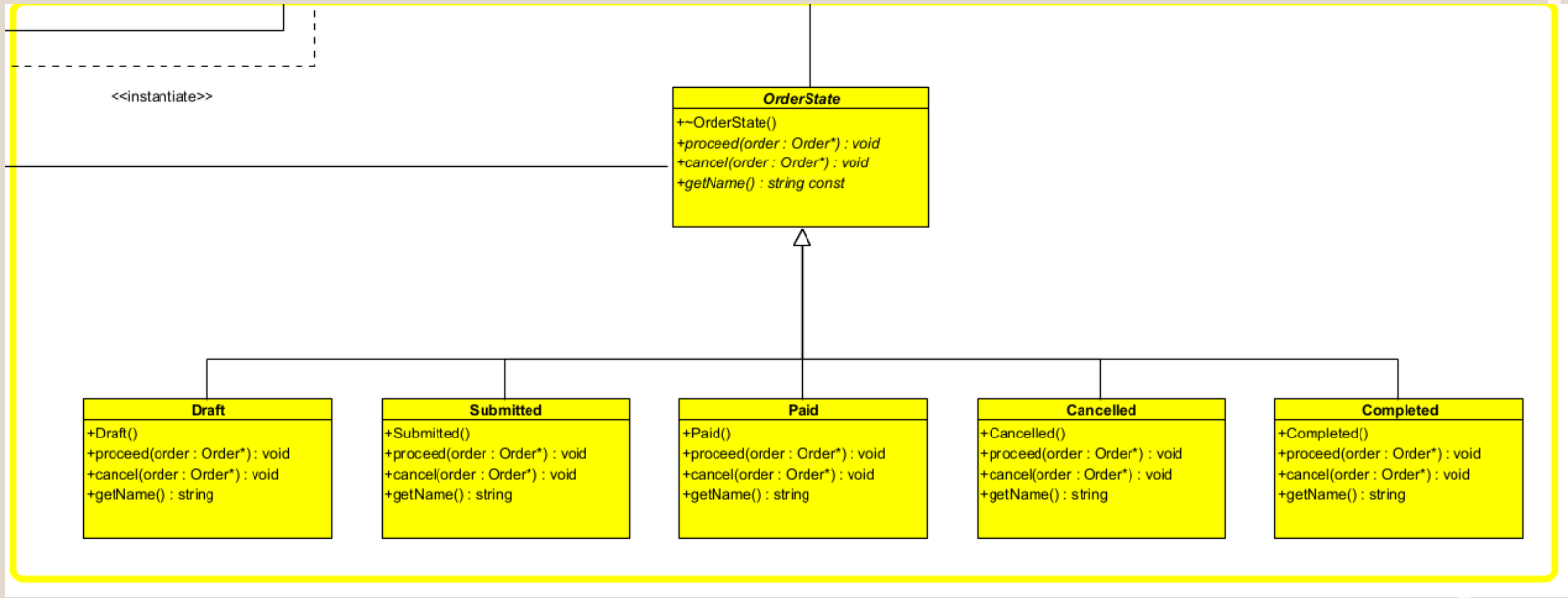
Participants:
Iterator: Iterator
concreteIterator: InventoryIterator, StockIterator, OrderIterator
Aggregate: Storage
concreteAggregates: Inventory, Stock, Order



State

Participants:
Context:Order
State: OrderState
concreteStates: Draft, Submitted, Paid, Cancelled, Completed

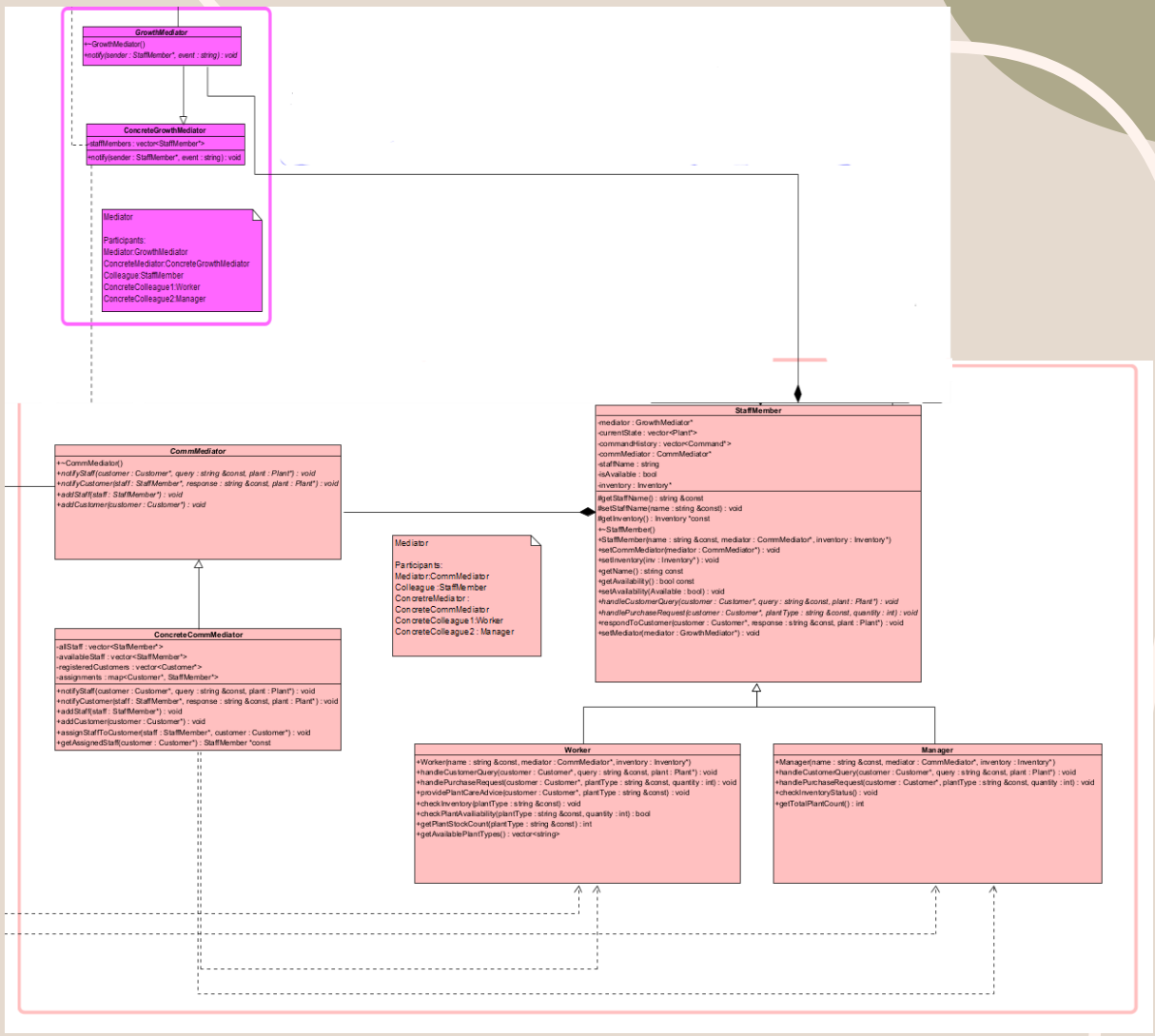




Mediator

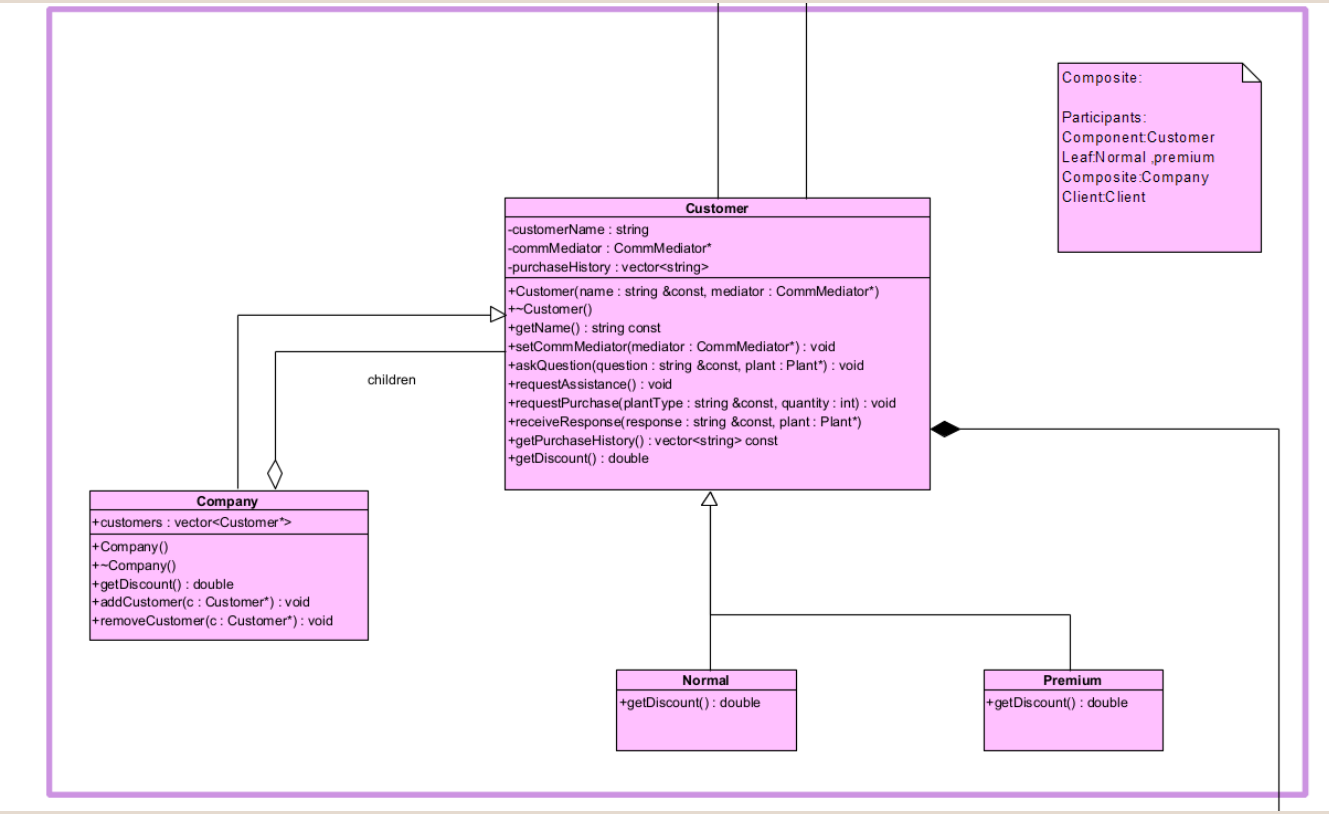
Participants:
Mediator: CommMediator
Colleague: StaffMember
ConcreteMediator: ConcreteCommMediator
ConcreteColleague1: Worker
ConcreteColleague2: Manager

Mediator: GrowthMediator
ConcreteMediator: ConcreteGrowthMediator
Colleague: StaffMember
ConcreteColleague1: Worker
ConcreteColleague2: Manager



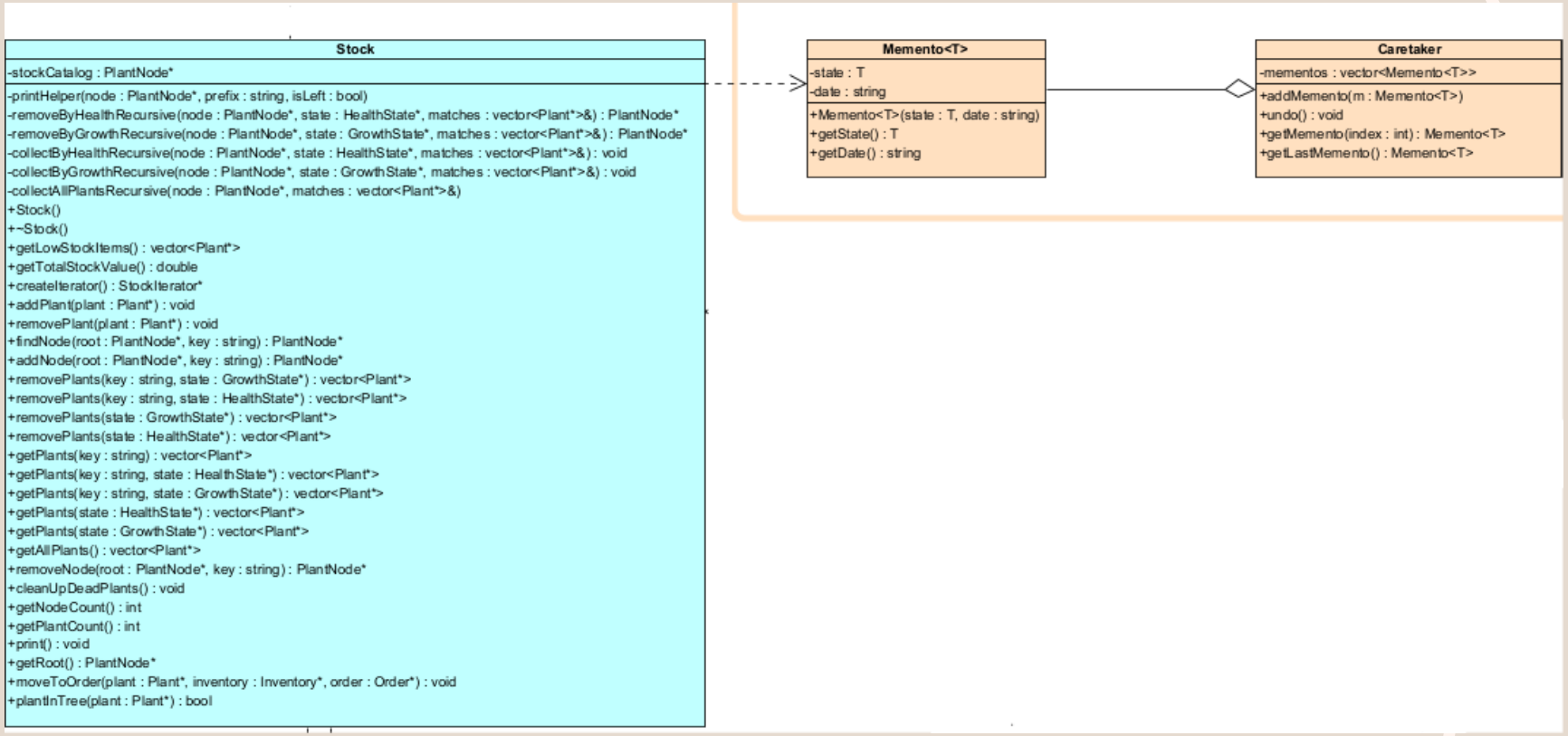
Composite

Participants:
Component: Customer
Leaf: Normal, Premium
Composite: Company
Client:Client



Memento

Participants:
Memento: Memento
CareTaker: Caretaker
Originator: Stock



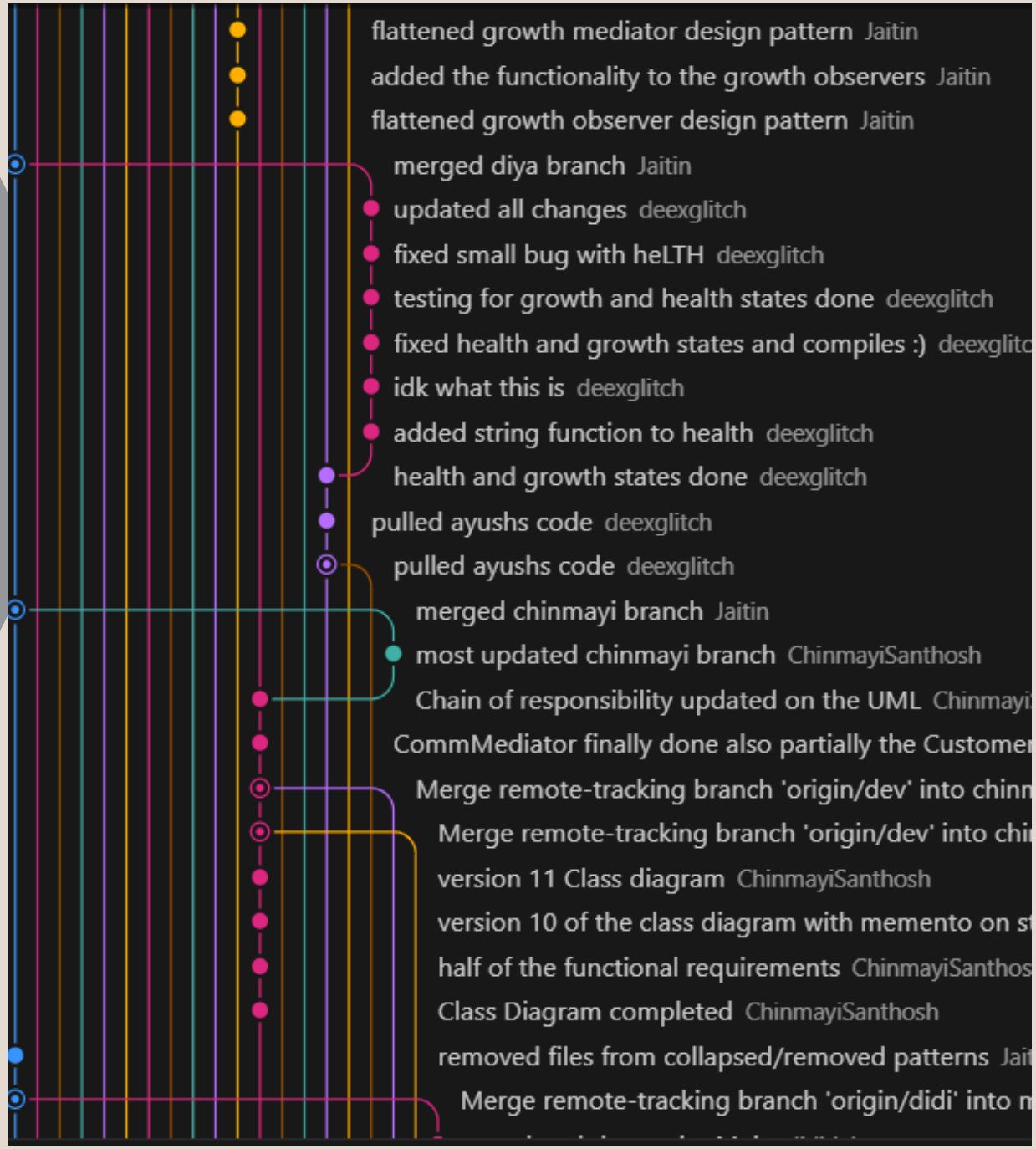
LETS DEMO

Memory Safety
Unit testing

```
All tests completed successfully!
==13383==
==13383== HEAP SUMMARY:
==13383==     in use at exit: 0 bytes in 0 blocks
==13383==   total heap usage: 1,123,046 allocs, 1,123,046 frees, 33,304,994 bytes allocated
==13383==
==13383== All heap blocks were freed -- no leaks are possible
==13383==
==13383== For lists of detected and suppressed errors, rerun with: -s
==13383== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

Git Strategy

We used the gitFlow branching strategy
Makes use of feature branches and multiple primary branches.
We then merged into demo main
And then the final merge into main branch



Commits

WHEN RUNNING COMMAND GIT SHORTLOG –SN

NOTE THAT SOME USERNAMES APPEAR TWICE

HENCE OUR FINAL TOTALS ARE:

JAITIN: 56

MAHADIO: 25

DIYA: 27

CHINMAYI:42

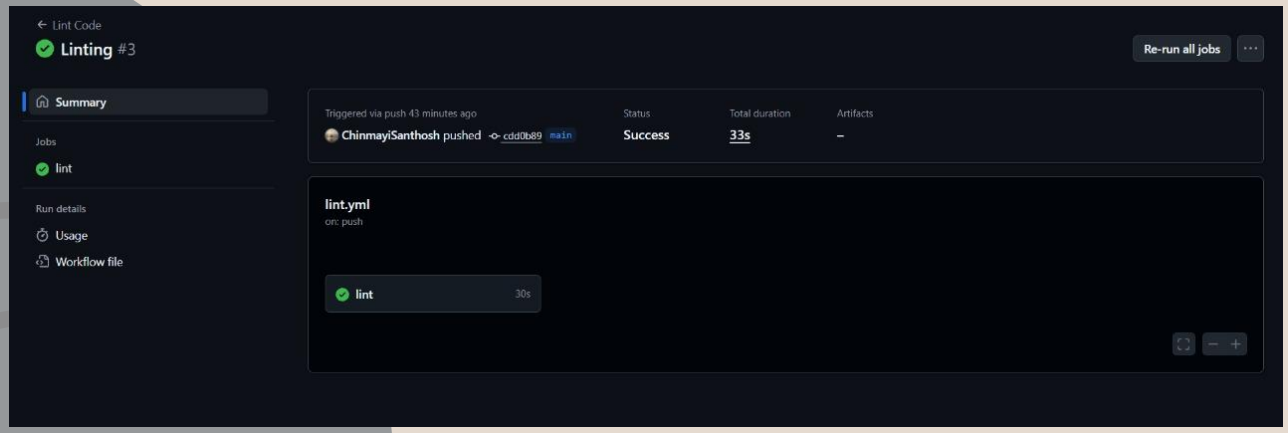
FABIO:25

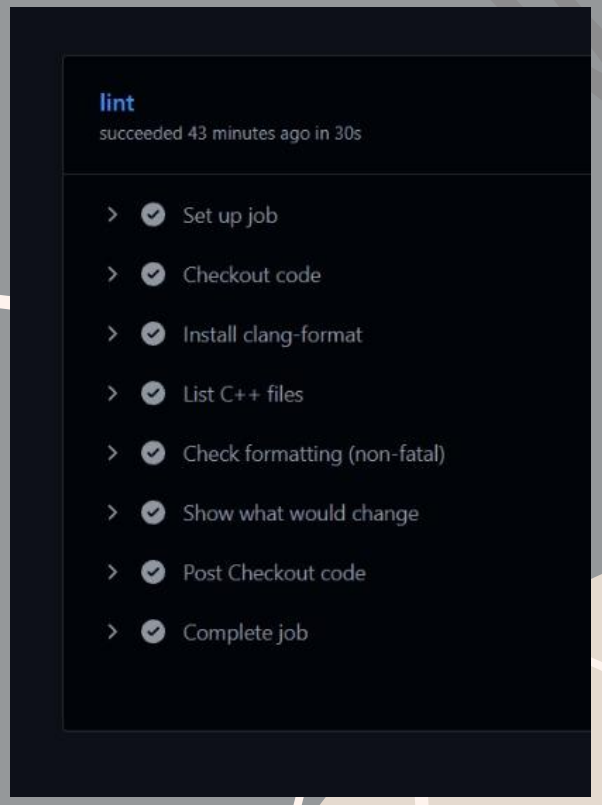
SHAVIR:25

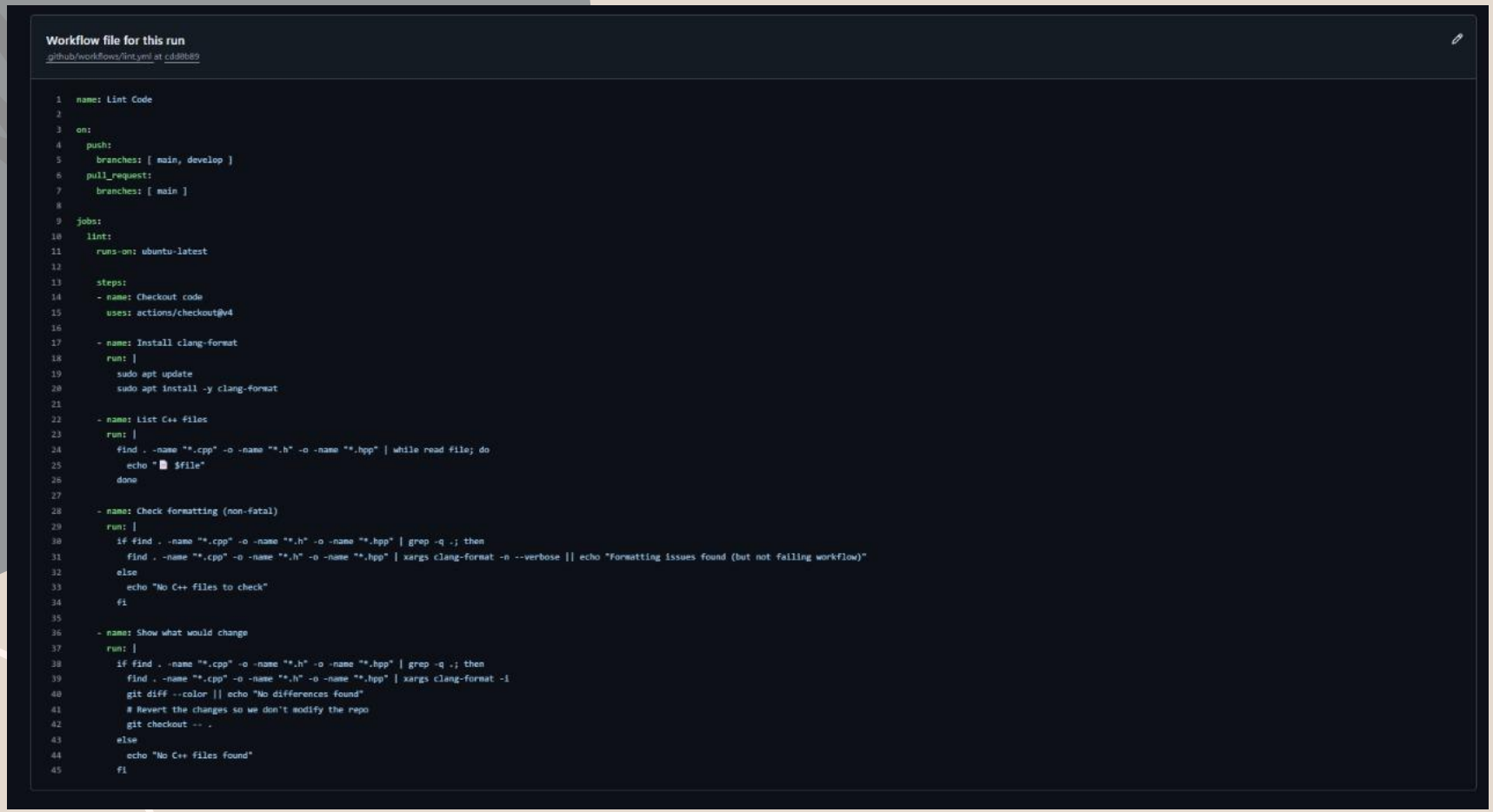
AYUSH: 68

```
diya@DESKTOP-FKIGHQ5:/mnt/c/Users/Diya/OneDrive -
r2/COS-214-Final-Project$ git shortlog -sn
    68  Ayush-B99
    56  Jaitin
    35  Ayush
    32  Chinmayi
    32  Life220
    25  diditlaka
    22  ShavirV
    17  Diya Narotam
    10  ChinmayiSanthosh
    10  Shavir Vallabh
    10  deexglitch
     1  Wave2055
```

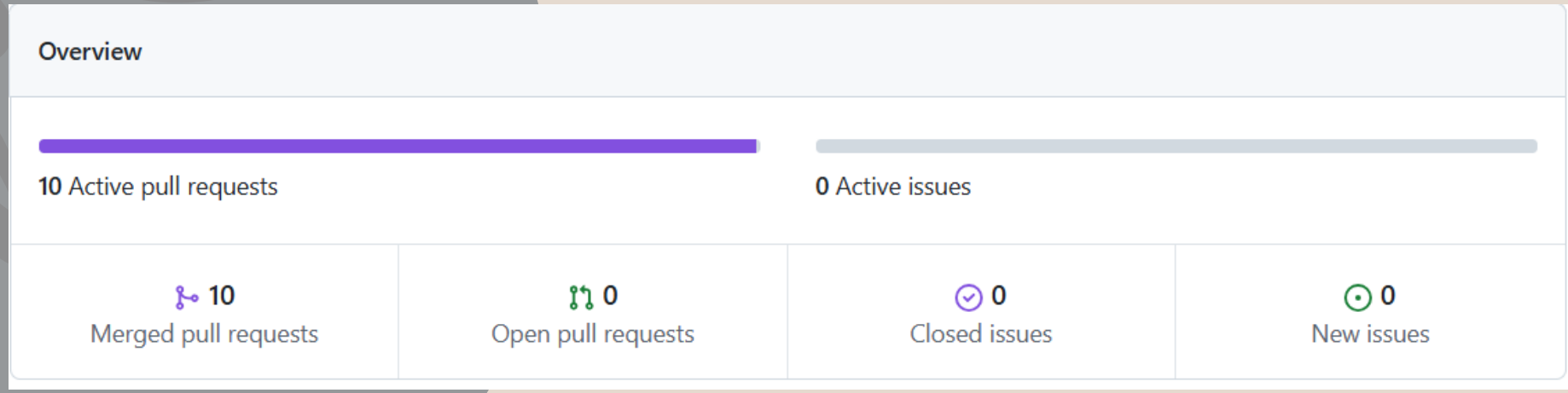
Linting


Lint Code
Linting #3
Re-run all jobs
Summary
Jobs
lint
Run details
Usage
Workflow file
Triggered via push 43 minutes ago
ChinmayiSanthosh pushed cdd0b89 main
Status
Success
Total duration
33s
Artifacts
–
lint.yml
on: push
lint
30s

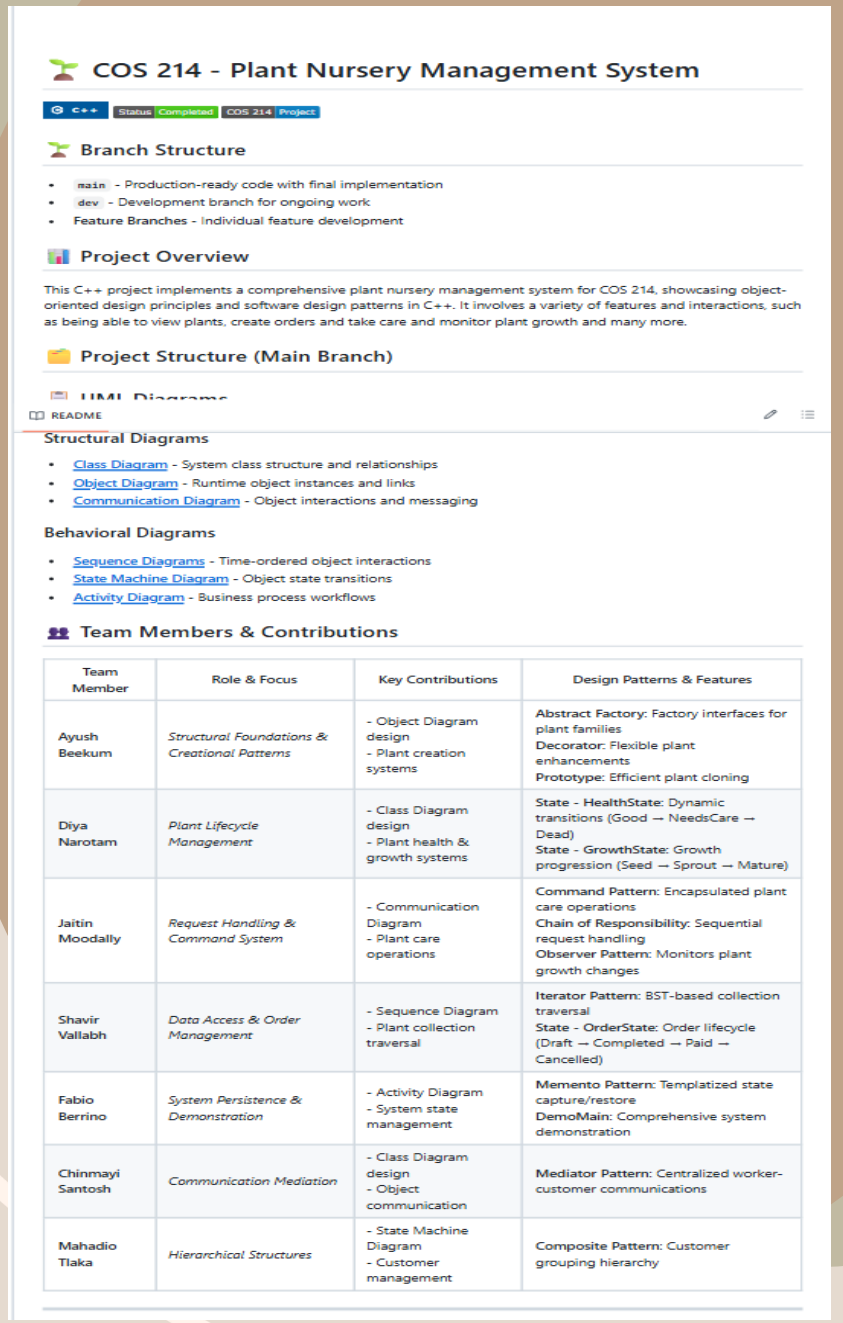

lint
succeeded 43 minutes ago in 30s
Set up job
Checkout code
Install clang-format
List C++ files
Check formatting (non-fatal)
Show what would change
Post Checkout code
Complete job


Workflow file for this run
.github/workflows/lint.yml at cdd0b89
name: Lint Code
on:
push:
branches: [main, develop]
pull_request:
branches: [main]
jobs:
lint:
runs-on: ubuntu-latest
steps:
- name: Checkout code
uses: actions/checkout@v4
- name: Install clang-format
run: |
sudo apt update
sudo apt install -y clang-format
- name: List C++ files
run: |
find . -name "*.cpp" -o -name "*.h" -o -name "*.hpp" | while read file; do
echo "📄 $file"
done
- name: Check formatting (non-fatal)
run: |
if find . -name "*.cpp" -o -name "*.h" -o -name "*.hpp" | grep -q .; then
find . -name "*.cpp" -o -name "*.h" -o -name "*.hpp" | xargs clang-format -n --verbose || echo "Formatting issues found (but not failing workflow)"
else
echo "No C++ files to check"
fi
- name: Show what would change
run: |
if find . -name "*.cpp" -o -name "*.h" -o -name "*.hpp" | grep -q .; then
find . -name "*.cpp" -o -name "*.h" -o -name "*.hpp" | xargs clang-format -i
git diff --color || echo "No differences found"
Revert the changes so we don't modify the repo
git checkout -- .
else
echo "No C++ files found"
fi

Pull requests



ReadMe



COS 214 - Plant Nursery Management System

C++ Status Completed COS 214 Project

Branch Structure

- `main` - Production-ready code with final implementation
- `dev` - Development branch for ongoing work
- **Feature Branches** - Individual feature development

Project Overview

This C++ project implements a comprehensive plant nursery management system for COS 214, showcasing object-oriented design principles and software design patterns in C++. It involves a variety of features and interactions, such as being able to view plants, create orders and take care and monitor plant growth and many more.

Project Structure (Main Branch)

README

Structural Diagrams

- Class Diagram - System class structure and relationships
- Object Diagram - Runtime object instances and links
- Communication Diagram - Object interactions and messaging

Behavioral Diagrams

- Sequence Diagrams - Time-ordered object interactions
- State Machine Diagram - Object state transitions
- Activity Diagram - Business process workflows

Team Members & Contributions

Team Member	Role & Focus	Key Contributions	Design Patterns & Features
Ayush Beekum	*Structural Foundations & Creational Patterns*	- Object Diagram design - Plant creation systems	**Abstract Factory:** Factory interfaces for plant families **Decorator:** Flexible plant enhancements **Prototype:** Efficient plant cloning
Diya Narotam	*Plant Lifecycle Management*	- Class Diagram design - Plant health & growth systems	**State - HealthState:** Dynamic transitions (Good → NeedsCare → Dead) **State - GrowthState:** Growth progression (Seed → Sprout → Mature)
Jaitin Moodally	*Request Handling & Command System*	- Communication Diagram - Plant care operations	**Command Pattern:** Encapsulated plant care operations **Chain of Responsibility:** Sequential request handling **Observer Pattern:** Monitors plant growth changes
Shavir Vallabh	*Data Access & Order Management*	- Sequence Diagram - Plant collection traversal	**Iterator Pattern:** BST-based collection traversal **State - OrderState:** Order lifecycle (Draft → Completed → Paid → Cancelled)
Fabio Berrino	*System Persistence & Demonstration*	- Activity Diagram - System state management	**Memento Pattern:** Templatized state capture/restore **DemoMain:** Comprehensive system demonstration
Chinmayi Santosh	*Communication Mediation*	- Class Diagram design - Object communication	**Mediator Pattern:** Centralized worker-customer communications
Mahadio Tlaka	*Hierarchical Structures*	- State Machine Diagram - Customer management	**Composite Pattern:** Customer grouping hierarchy

Doxygen

File List

Here is a list of all documented files with brief descriptions:

[detail level 1 2]

include	
AloeVera.h	Concrete product representing an Aloe Vera succulent plant (medium size)
BirdOfParadise.h	Concrete product representing a Bird of Paradise tropical plant (medium size)
Cancelled.h	Concrete OrderState representing an order that has been cancelled
Caretaker.h	Implements the Caretaker role in the Memento design pattern
CarnivorousPlantFactory.h	Concrete factory for creating carnivorous plants
Client.h	
Command.h	Abstract command interface for plant care operations
CommMediator.h	Abstract mediator interface for customer-staff communication
Company.h	Composite component representing corporate customer groups
Completed.h	Concrete OrderState representing an order that has been completed
ConcreteCommMediator.h	Concrete implementation of the communication mediator
ConcreteGrowthObserver.h	Observer that monitors plant growth changes and coordinates care
Condelabra.h	Concrete product representing a Condelabra succulent plant (large size)
Customer.h	Abstract component in the Composite pattern for customer hierarchy
Daisy.h	Concrete product representing a Daisy temperate plant (small size)
Dead.h	Concrete state representing a deceased plant
Draft.h	Concrete OrderState representing an order in draft form
Fertilizer.h	Concrete command for fertilizing plants
FertilizerDecorator.h	Concrete decorator for adding fertilizer features to plants
FertilizerHandler.h	Concrete handler for fertilizer-related plant care requests
Good.h	Concrete state representing a healthy plant
GreenHouse.h	Abstract factory for creating plants of different sizes

Questions, comments & concerns

thank you

MAHADIO

SHAVIR

DIYA

AYUSH

CHINMAYI

JAITIN

FABIO